Human Physiology · PHYS-UG-MEMB-01

PHYS-MEMB-1 Membrane Potentials and the Action Potential

Work each activity on blank paper with your notes closed, then open the key and mark what you missed. Do the two brain dumps for one competency on separate days instead of back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
PHYS-UG-MEMB-01-C01				
PHYS-UG-MEMB-01-C02				
PHYS-UG-MEMB-01-C03				
PHYS-UG-MEMB-01-C04				
PHYS-UG-MEMB-01-C05				
PHYS-UG-MEMB-01-C06				

Brain dump 1 · foundational retrieval PHYS-UG-MEMB-01-C01-BD1

Draw the resting membrane and label what sets it

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C01 Activity: PHYS-UG-MEMB-01-C01-BD1 Date: _____

Without looking at your notes, draw a section of a neuron membrane with the inside on one side and the outside on the other. Write the resting membrane potential of a typical neuron in millivolts and say what the minus sign in front of that number means. Place the four ions that matter here, sodium, potassium, chloride and the negatively charged proteins, on the side where each one is more concentrated. Then write three short reasons the inside sits negative at rest: which ion the resting membrane leaks most easily and which channel lets it out, which ion is kept out because few of its channels are open at rest, and which particles are stuck inside because they are too large to cross. Finish by defining resting membrane potential in one sentence a classmate would understand.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application PHYS-UG-MEMB-01-C01-BD2

Move the resting potential and say which way

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C01 Activity: PHYS-UG-MEMB-01-C01-BD2 Date: _____

A neuron is sitting at its normal resting potential. One at a time, work through these four changes and for each one write whether the membrane potential moves closer to zero or further below zero, name the direction of that move using the words depolarization or hyperpolarization, and give the reason in one sentence. First, the potassium concentration outside the cell rises. Second, extra potassium leak channels open. Third, a few sodium channels open while the cell is still at rest. Fourth, the fluid around the cell is replaced with a solution that has no sodium in it at all. For the fourth one, also write whether you would expect the resting potential to change much and explain your answer.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval PHYS-UG-MEMB-01-C02-BD1

Draw the sodium potassium pump at work

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C02 Activity: PHYS-UG-MEMB-01-C02-BD1 Date: _____

Draw the sodium potassium pump in a membrane. Write how many sodium ions it moves and in which direction, how many potassium ions it moves and in which direction, and what molecule it spends to do it. Write the net charge the pump moves out of the cell with each cycle and say in one sentence how that small charge movement adds to the negative resting potential. Then write the approximate concentration of sodium and of potassium inside the cell and outside the cell, and beside each pair draw an arrow showing which way that ion would drift if a channel for it opened. Last, define the term concentration gradient in your own words and say which structure builds and maintains these two gradients.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application PHYS-UG-MEMB-01-C02-BD2

Lose the pump and follow the consequences

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C02 Activity: PHYS-UG-MEMB-01-C02-BD2 Date: _____

A cell is poisoned so its sodium potassium pumps stop working, but every channel in the membrane still works normally. Write what happens over the next several minutes to the sodium concentration inside the cell, the potassium concentration inside the cell, the size of each gradient, and the resting membrane potential. Then write whether the cell can still fire one action potential right after the pump stops and explain why, and whether it can still fire them an hour later and explain why. Finish with two sentences on why physiologists call the gradients a stored supply of energy for the cell.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

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What needs more practice?

Next retrieval date: _____

Prediction exercise PHYS-UG-MEMB-01-C02-PR01

Predict what rising outside potassium does

The potassium concentration in the fluid around a neuron climbs from its normal value to well above normal. For each item below, write whether it increases, decreases or stays the same, then write one sentence of reasoning that names the gradient or the channel involved. Do not look anything up until you have written all five.

What changes	Up, down, or no change	Why, in one or two lines
The size of the potassium concentration gradient across the membrane		
The rate at which potassium leaks out of the cell at rest		
The resting membrane potential, stated as closer to zero or further below zero		
How close the membrane sits to the threshold voltage		
The number of voltage gated sodium channels that are able to open when a stimulus arrives		

Confidence before checking

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After checking my notes

What did I remember correctly?

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What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval PHYS-UG-MEMB-01-C03-BD1

Two kinds of electrical signal, side by side

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C03 Activity: PHYS-UG-MEMB-01-C03-BD1 Date: _____

Draw two sets of axes with voltage on the vertical axis and time on the horizontal axis. On the first, sketch a graded potential, meaning a small local change in membrane voltage. On the second, sketch an action potential, meaning the large all or nothing signal a neuron sends down its axon. Beside each drawing write where on the neuron it normally happens, what opens the channels that cause it, roughly how large it is in millivolts, how far it travels before it fades, and whether its size can vary with the strength of the stimulus. Then define threshold in one sentence and mark it on the second drawing. Last, write what the phrase all or nothing means and why it applies to only one of these two signals.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application PHYS-UG-MEMB-01-C03-BD2

From small inputs to a fired axon

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C03 Activity: PHYS-UG-MEMB-01-C03-BD2 Date: _____

Three small depolarizing graded potentials arrive at one neuron. Write what happens at the start of the axon in each of these situations and give your reason each time. First, the three arrive one after another with long gaps between them. Second, the three arrive at nearly the same moment at the same spot on the dendrite. Third, the three arrive at the same moment but at three spots far apart on the dendrites. Fourth, two depolarizing graded potentials and one hyperpolarizing graded potential arrive together. Then answer this: a stimulus is doubled in strength and the neuron is already firing. Write what changes in the signal the axon sends, since the height of each action potential cannot change, and name the feature of the signal that carries stimulus strength instead.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table PHYS-UG-MEMB-01-C03-CP01

Graded potential compared with action potential

Fill in every cell of the table from memory, in short phrases rather than sentences. Leave no cell blank; if you are unsure, write your best answer and circle it so you can check it against the key.

Feature	Graded potential	Action potential
Where on the neuron it usually occurs		
What causes the channels to open		
Main ion or ions that move		
Typical size of the voltage change		
Does size vary with stimulus strength		
How far it travels and whether it fades		
Does it need to reach threshold		
Is there a refractory period		
How stimulus strength is coded		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval PHYS-UG-MEMB-01-C04-BD1

Draw the action potential and label every gate

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C04 Activity: PHYS-UG-MEMB-01-C04-BD1 Date: _____

Draw one action potential with voltage on the vertical axis and time on the horizontal axis. Mark the resting potential, the threshold voltage, the peak voltage, and the lowest point of the undershoot, and write a number in millivolts beside each one. Label these parts on your drawing: the graded depolarization up to threshold, the rising phase, the peak, the falling phase, and the undershoot, also called hyperpolarization. Beside each part write which channel is opening or closing at that moment, which ion is moving, and which direction that ion is going. For the voltage gated sodium channel, draw both of its gates, name them, and write what each gate is doing at rest, at threshold, at the peak, and during the falling phase. Then write why the potassium channels are described as slow and what that slowness produces at the end of the tracing.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application PHYS-UG-MEMB-01-C04-BD2

Change one gate and redraw the tracing

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C04 Activity: PHYS-UG-MEMB-01-C04-BD2 Date: _____

Redraw the normal action potential lightly, then on top of it sketch what the tracing looks like in each of these four situations and write one sentence saying why. First, a toxin holds the inactivation gates of the sodium channels open so they never close. Second, a drug slows the voltage gated potassium channels so they open later than normal. Third, half of the voltage gated sodium channels are blocked and cannot open at all. Fourth, the sodium concentration outside the cell is lowered to half of normal. For each one, state what happens to the peak voltage, the length of the action potential, and whether the cell still reaches threshold.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

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What needs more practice?

Next retrieval date: _____

Retrieval exercise PHYS-UG-MEMB-01-C04-RT01

Rebuild the action potential step by step

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C04 Activity: PHYS-UG-MEMB-01-C04-RT01 Date: _____

Rebuild the full sequence of the action potential from memory as a numbered list, one event per line and in order, starting with a stimulus that depolarizes the membrane to threshold and ending with the membrane back at its resting potential. Write at least eight steps. Beside each step name the channel involved, the ion moving, and the direction it moves. Do not draw a graph here; write the sequence in words so you can see whether the order is solid in your memory.

Rebuild from memory: The sequence of channel events in one action potential

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application PHYS-UG-MEMB-01-C04-CA01

A numb tooth after a dental injection

A healthy 20 year old student has a cavity filled. Before drilling, the dentist injects lidocaine, a local anesthetic, into the gum beside the tooth. Lidocaine crosses the membranes of nearby neurons and binds inside the pore of voltage gated sodium channels, holding them closed. Within a few minutes the student feels nothing sharp in that tooth or in the nearby lip, although she can still move her jaw normally and can still feel firm pressure when the dentist presses on the gum. About two hours later the numbness wears off completely with no lasting damage.

1. Recognize. Name which part of the electrical signal the drug stops and state whether the problem is in the resting potential, the graded potential, or the action potential.

2. Explain. Write the mechanism in three or four sentences: what the blocked channel can no longer do, which phase of the action potential disappears, and why the signal therefore never travels along the axon to the brain.

3. Differentiate. Explain why pain fades first while pressure sensation and jaw movement are still present, using axon diameter and myelin to support your answer.

4. Investigate. Write two observations you could make at the chair, without any equipment, that would tell you whether the block has taken effect in sensory axons, and say what each observation shows.

5. Decide. Decide whether the resting membrane potential of the blocked neuron is still normal and state your answer as a single sentence.

6. Justify. Justify that decision by naming which channels and which pump set the resting potential and explaining why a drug that blocks only voltage gated sodium channels leaves it unchanged.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval PHYS-UG-MEMB-01-C05-BD1

The two refractory periods on one tracing

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C05 Activity: PHYS-UG-MEMB-01-C05-BD1 Date: _____

Draw an action potential and mark the absolute refractory period and the relative refractory period along the time axis. Define each one in a single sentence, where absolute means no second action potential is possible no matter how strong the stimulus and relative means a second one is possible but needs a stronger stimulus than usual. Beside the absolute refractory period write the state of both sodium channel gates and say which gate is responsible. Beside the relative refractory period write what the slow potassium channels are still doing and why that makes threshold harder to reach. Then write what the sodium channels must do before the cell is fully ready again, and name the voltage range the membrane has to return to for that to happen.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application PHYS-UG-MEMB-01-C05-BD2

Why the signal runs one way and has a speed limit

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C05 Activity: PHYS-UG-MEMB-01-C05-BD2 Date: _____

An action potential is halfway along an axon. The patch of membrane just behind it has only just finished firing. Write what state the sodium channels in that patch are in and use it to explain why the signal keeps moving forward instead of spreading backward. Next, a researcher stimulates a neuron faster and faster. Write what sets the maximum number of action potentials that neuron can fire in one second and estimate that number if the absolute refractory period lasts about one millisecond. Last, write two sentences on what would go wrong in a muscle or a nerve if refractory periods did not exist at all.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval PHYS-UG-MEMB-01-C06-BD1

Draw a myelinated axon and show how the signal jumps

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C06 Activity: PHYS-UG-MEMB-01-C06-BD1 Date: _____

Draw a short length of myelinated axon. Label the myelin sheath, one node of Ranvier, meaning a bare gap in the sheath, and the cell that makes myelin in the peripheral nervous system and the one that makes it in the central nervous system. Write what myelin is made of and the two things it does to the axon membrane underneath it. Mark where the voltage gated sodium channels are concentrated. Then show with arrows how the action potential moves from node to node, name that pattern, and write a one sentence definition of it. Finish by listing the two factors that make an axon conduct faster and write the approximate conduction speed of a large myelinated axon and of a small unmyelinated one.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application PHYS-UG-MEMB-01-C06-BD2

Strip the myelin and compare the cost

Course: Human Physiology Subject: Human Physiology Chapter and topic: PHYS-MEMB-1 Membrane Potentials and the Action Potential Competency: PHYS-UG-MEMB-01-C06 Activity: PHYS-UG-MEMB-01-C06-BD2 Date: _____

An axon loses patches of its myelin while the axon itself stays intact. Write what happens to conduction speed, to whether the signal reaches the end of the axon at all, and to the sensation or movement that axon serves. Explain in two or three sentences why losing the sheath can stop a signal completely rather than only slowing it, using what you know about where the sodium channels sit in a myelinated axon. Then compare two ways a body could build a fast axon, by adding myelin or by making the axon much wider, and write which one costs less space in a nerve and why. Finish by naming the type of axon that carries sharp pain and the type that carries dull aching pain, and say which one is myelinated.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

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What needs more practice?

Next retrieval date: _____

Board-style question PHYS-UG-MEMB-01-C05-Q01

During the absolute refractory period of a neuron, a second stimulus of any strength cannot produce another action potential. Which statement best explains why?

A. The sodium potassium pump has not yet restored the sodium and potassium concentration gradients

B. The membrane potential is more negative than the resting potential, so threshold is further away

C. The inactivation gates of the voltage gated sodium channels are closed and will not reopen until the membrane repolarizes

D. All of the voltage gated potassium channels are closed, so potassium cannot leave the cell

Why did you pick it?

Board-style question PHYS-UG-MEMB-01-C06-Q02

A myelinated axon conducts an action potential far faster than an unmyelinated axon of the same diameter. Which feature of the myelinated axon accounts for most of that difference?

A. Myelin supplies extra ATP to the sodium potassium pumps along the axon

B. The sheath keeps charge from leaking across the membrane, so the depolarization spreads quickly to the next node of Ranvier, where the sodium channels are clustered

C. Myelin raises the number of sodium channels spread evenly along the whole length of the axon membrane

D. The sheath shortens the absolute refractory period so each patch of membrane can fire again sooner

Why did you pick it?